

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS
Conditional Probability and Bayes Theorem

Week 3 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



Last week we calculated two numbers in real data:

$$P(\text{positive} | \text{diseased}) = \frac{154}{208} = 0,7404$$

$$P(\text{diseased} | \text{positive}) = \frac{154}{1115} = 0,1384$$

The picture is the same. The denominator is different. The result is more than five times different.

A common mistake But most of the time we only have one direction.

The doctor knows $P(\text{positive} | \text{diseased})$ — he has tested the test on patients.

And the patient needs $P(\text{diseased} | \text{positive})$.

This week's question How do we calculate the other if we know one direction?

Learning Objectives for Week 3

By the end of this week, you will:

1. Illustrate conditional probability with a tree diagram and apply the multiplication rule in chain form.
2. Define the concept of division and apply the complete probability formula.
3. Derive and apply Bayes' theorem.
4. Explain the effect of the base rate and determine the base rate error.
5. Solve diagnostic testing and credit scoring problems.
6. Recognize and correct a prosecutorial error.

Learning Outcome CLO2. Apply the rules of probability, including conditional probability and Bayes' theorem, to economic problems (Bloom: Application and Analysis).

Today's strong point. We will test Bayes' theorem on real data: the answer given by the formula should be exactly the same as the answer given by direct counting. If not, we have made a mistake.

Tree diagram and multiplication rule

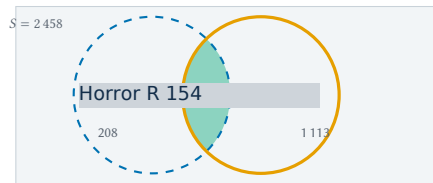
Conditional probability: narrowing the space

Definition

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) > 0$$

The news that B has happened narrows the entire space to B . Now B is the new “world”.

A common mistake When calculating conditional probability, the denominator changes. Students often change the picture and forget the denominator.



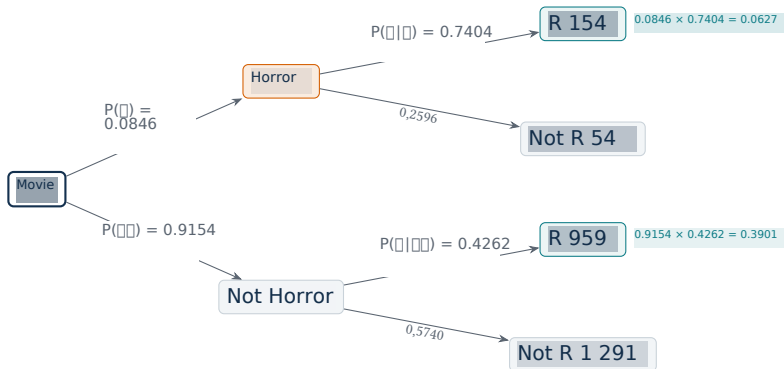
$$P(R|Horror) = \frac{154}{1113}$$

208 denominator = Horror

$$P(Horror|R) = \frac{154}{208}$$

1113 denominator = R

Tree Diagram: The Language of Conditional Probability



Tree rules. Multiply along a branch; add along branches. The sum of the probabilities from each node is 1.

Audience question $0.0627 + 0.3901 = 0.4528$. What number is this? Why did it come out that way?

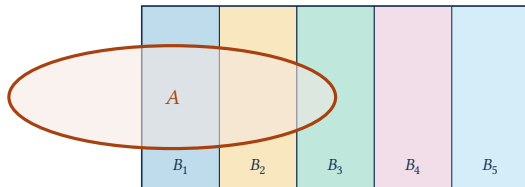
Complete probability formula

Partition

Definition The events $B_1, B_2, \dots, B_k$ are called a partition of Ω if:

- ▶ they do not intersect in pairs: $B_i \cap B_j = \emptyset$
- ▶ the union is the entire space: $\bigcup_{i=1}^k B_i = \Omega$

That is, each outcome falls into exactly one B_i .



Ω is divided into partitions: $\Omega = \bigcup_{i=1}^k (B_i \cap \Omega)$

Complete probability formula

$$P(A) = \sum_{i=1}^k P(A|B_i) P(B_i)$$

Proof. $B_i \cap B_j$ do not intersect in pairs, their union is Ω . According to axiom A3, the probabilities are added; we apply the multiplication rule to each term. $\square$

In real data: genres form a division

Genre	n	$P(A)$	$P(B A)$	$P(A B)$	$P(B)$
Action	358	0.1456	174	0.4860	0.0708
Adventure	363	0.1477	14	0.0386	0.0057
Comedy	600	0.2441	292	0.4867	0.1188
Drama	929	0.3779	479	0.5156	0.1949
Horror	208	0.0846	154	0.7404	0.0627

Total 2,458 1.0000 1 113 — 0.4528

Check. Direct calculation:

$$P(B) = \frac{1113}{2458} = 0.452807$$

Complete probability formula: 0.452807. ✓

Difference = 0 — not a rounding error either.

Common error Check the divisibility condition. $358 + 363 + 600 + 929 + 208 = 2458$. If the sum does not match — it is not a divisor, the formula does not work. This is the most common error.

Bayes' theorem

Bayes' theorem: trinomial induction

Step 1. Definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Step 2. Image multiplication rule:

$$P(A \cap B) = P(A|B) P(B)$$

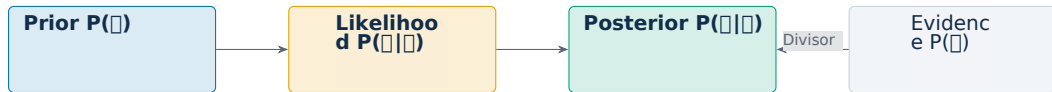
Step 3. Formula for total probability in denominator:

$$P(A) = \sum_i P(A|B_i) P(B_i)$$

Bayes' theorem

$$P(B_i|A) = \frac{P(A|B_i) P(B_i)}{\sum_j P(A|B_j) P(B_j)}$$

Bayes's theorem is not a new rule. It is a combination of the definition, the multiplication rule, and the formula for total probability. It doesn't have to be memorized; it has to be derived.



Prior Confidence before seeing evidence. Often the base rate is a proportion of the population. Confidence How likely is it to see this evidence if $\square$ is true. The posterior confidence after considering the evidence is the number we need.

Bayes = confidence update. Prior probability $\times$ strength of evidence = posterior probability.

Testing Bayes on real data A movie is rated R.
What is the probability that it is a Horror?

By Bayes:

$$\begin{aligned} P(H|R) &= \frac{P(R|H) P(H)}{P(R)} \\ &= \frac{0.7404 \times 0.0846}{0.4528} = \frac{0.0627}{0.4528} = \mathbf{0.1384} \end{aligned}$$

By direct calculation:

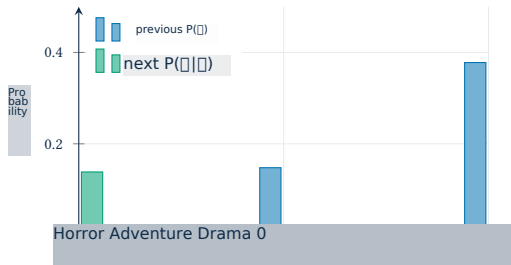
$$P(H|R) = \frac{154}{1113} = \mathbf{0.1384} \quad \checkmark$$

Why is this important? There were two ways here — that's why we were able to test the formula.

In real problems, direct calculation is not possible: there is no patient list. Therefore, we trust the formula — but first we tested it in a verified place.

The common error is $0.7404 \rightarrow 0.1384$. Not two identical movies — the same 154 movies, two different denominators.

How does evidence change confidence? For three genres: previous → next



Genre previous next coeff.

Horror 0.0846 0.1384 $\times 1.64$
Adventure 0.1477 0.0126 $\times 0.09$
Drama 0.3779 0.4304 $\times 1.14$

The evidence works both ways. An R rating increases the probability of a Horror, but decreases the probability of an Adventure by a factor of 12. Adventure films almost never get an R rating.

Base rate

Classic problem: screening test

Sensitivity of the test: $P(+ | \square) = 0.99$.

Specificity of the test: $P(- | \square\square) = 0.95$, so $P(+ | \square\square) = 0.05$.

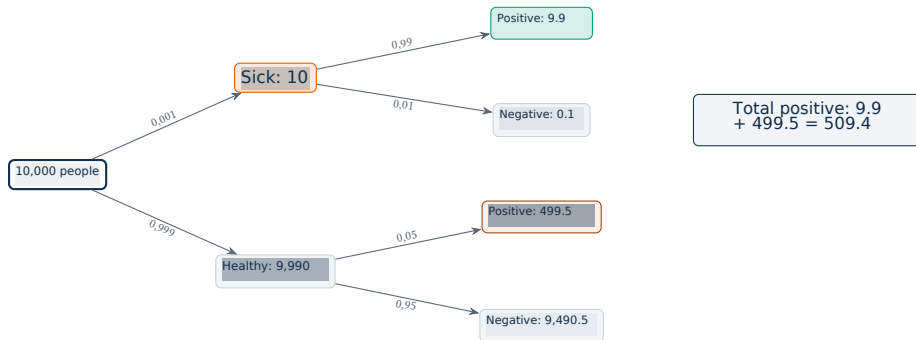
Audience question Your test came back positive. What is the probability that you have the disease? Write down your answer.

Common mistake What do most people say? The most common answers in surveys of health care workers are 95% or 99%.

That is, people use $P(+ | \square)$ instead of $P(\square | +)$. This is a fallacy of that direction.

The correct answer is on the next two slides.

Solving with frequencies: 10,000 people



$$P(\square|+) = \frac{9.9}{509.4} = 0.0194$$

A common mistake is that 98% of those who get a positive result are healthy: there are so many healthy people that 5% of them are more than all the sick people.

Same result with Bayes' formula

Denominator — complete probability formula:

$$\begin{aligned}P(+)&= P(+|\square)P(\square) + P(+|\square\square)P(\square\square) \\&= 0,99(0,001) + 0,05(0,999) = 0,05094\end{aligned}$$

Bayes:

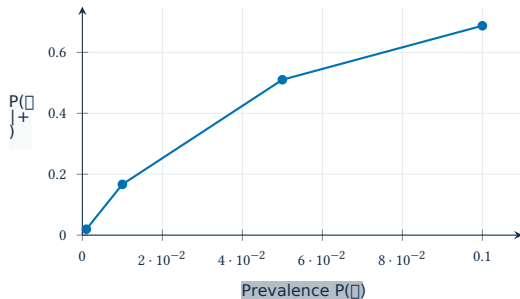
$$\begin{aligned}P(\square|+) &= \frac{0,99 \times 0,001}{0,05094} = \frac{0,00099}{0,05094} = \mathbf{0,0194}\end{aligned}$$

For a negative result: $P(\square\square|-) = 0,999989$.

Two ways — one answer. Solving with frequencies is more intuitive; solving with a formula is more general. You should know both on the exam.

Practical advice If you get confused about a Bayes problem, draw a table of 10,000 people. This method almost never cheats.

It all depends on the base rate



$P(+)$	$P(+ +)$
0,001	1,94%
0,01	16,67%
0,05	51,03%
0,10	68,75%

Common Error The test has not changed. Both the sensitivity and specificity are the same. Only the base rate has changed — and the result has increased from 1.94% to 68.75%.

This is why mass screening is selective. Testing the entire population for a rare disease would result in a flood of false positives.

Base Rate Fallacy

Definition Base Rate Fallacy — focusing on a clear and vivid piece of evidence and ignoring the overall frequency of an event (the base rate).

A common fallacy Why is this a natural fallacy? The numbers we are given (0.99, 0.95) are clear and obvious. The base rate (0.001) seems like “background information.” In fact, it is the most important number.

Countermeasure: frequency form. Instead of “ $P(+ | \square) = 0.05$,” say “499 out of 10,000 healthy people test positive.”

Research shows that people are significantly better at solving the same problem when presented in frequency form.

Audience Question Where does the base rate fallacy occur? Name at least two areas.

**Economic
application**

Credit scoring model

Hypothetical bank model The default rate in the portfolio is $P(\square) = 0.03$.

The model identifies 90% of defaulters:
 $P(\text{flag} | \square) = 0.90$.

It falsely flags 10% of good customers:
 $P(\text{flag} | \square\square) = 0.10$.

$$P(\text{flag}) = 0.90(0.03) + 0.10(0.97) = 0.1240$$

$$P(\square | \text{flag}) = \frac{0.027}{0.1240} = 0.2177$$

A common mistake 78.2% of flagged applications do not actually default. Even though the model is advertised as "90% accurate."

The other side:

$$P(\square | \text{no flag}) = \frac{0.003}{0.8760} = 0.0034$$

Only 0.34% of applications without a flag default — the model is very reliable in rejecting.

What decision does this number require?

Audience Question Out of 100 flagged applications, 22 default, and 78 are good customers.

(a) What does the bank lose if it rejects all flagged applications?

(b) What additional information would improve the quality of the decision?

(c) Is this fair to these 78 customers?

Statistics do not make decisions. They only count the consequences. A rejection decision requires a comparison of costs and benefits—this will be discussed in Week 10, when we discuss Type I and Type II errors.

A common mistake Ethical. If the base rate differs across groups, the same model will affect different groups differently. This is a question of algorithmic fairness.

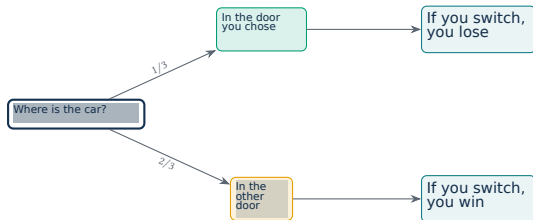
Monty Hall

Monty Hall Problem

Three doors: one with a car, two with a goat.

1. You choose one door.
2. The presenter opens the other two with a goat (he knows).
3. Do you switch the choice?

Audience question Should you switch? Is there a difference?



Probability of winning a switch = $2/3$. Simulation (100,000 games): stay 0.3326, switch 0.6674.

Common mistake Key condition: the initiator knows and always reveals the goat. If it opens by chance — the answer is $1/2$.

**Summar
y**

A common fallacy Prosecutor's fallacy. In court: "The probability of the evidence matching by chance is one in a million. Therefore, the probability of the accused being innocent is one in a million."

This equates two completely different numbers:

$$P(\text{evidence} | \text{innocent}) \neq P(\text{innocent} | \text{evidence})$$

Bayes shows that the latter also depends on the base rate—that is, the size of the pool of suspects.

With numbers. The population of the city is 2 million; the evidence matches by chance in one person in a million.

The guilty one is definitely a match: 1 person. The expected number of matches from the remaining 2,000,000 – 1 people: ≈ 2 people. Total matches: ≈ 3 .

$$P(\text{guilty} | \text{fit}) \approx \frac{1}{3} = 0,33$$

Not "one in a million" — one in three.

A common mistake This mistake has caused wrongful convictions in real court cases. The price of statistical error is human fate.

Common Student Mistakes and Misconceptions

Misconception Why it's wrong

"The test is 99% accurate — so if I get a positive result, I'm 99% sick." This is reading $P(+ | \square)$ as $P(\square | +)$. If the base rate is 0.001, the answer is 1.94%.

"The base rate is just background information." On the contrary, it's the most impressive number. If the prevalence changes from 0.001 to 0.10, the answer goes from 1.94% to 68.75%.

"The formula for absolute probability can be applied to any number of events." Only to the division. Check the sum: $358+363+600+929+208 = 2,458$.

"Bayes's theorem is a unique formula to remember." It is a combination of the definition + multiplication rule + total probability formula. Derive it.

"The next probability is always greater than the previous one." For Adventure, $0.1477 \rightarrow 0.0126$: the proof has reduced the confidence by a factor of 12.

"There are two doors left in Monty Hall — so it's 50/50." The starter does not open them at random. His knowledge adds information; substitution gives $2/3$.

"The model is 90% accurate — so 90% of the flags are correct." $P(\square | \text{flag}) = 0.2177$. 78.2% of the flags are false positives.

Key conclusions

1. A tree diagram is the most reliable tool for conditional probability problems: multiply along a branch, add along a branch.

2. The complete probability formula applies only to division—always check the sum.

3. Bayes' theorem is not a new rule; it is a combination of three well-known rules.

4. The subsequent probability depends on both the base rate and the strength of the evidence. It is a mistake to omit either.

5. If you get confused, draw a frequency table for 10,000 people.

6. Confusing $P(A|B)$ and $P(B|A)$ can cause real harm in medical, legal, and credit situations.

Next week Week 4 — Random variables and discrete distributions. We move from events to numbers: probability mass function, expectation, variance; Bernoulli, binomial, and Poisson distributions.

This week's assignments

Assignment Content Due Date

CONTROL WORK 1 Graded. Weeks 1-3, 45 minutes this week Required reading Newbold Chapter 3; McClave Chapter 3; Heumann Chapter 6 Before Lecture 1 Lab W03-lab — Bayes, trees, Monty Hall In the lab Homework W03-homework — 6 problems, 40 points Week 4, Monday Problem set 2 MT2 due this week Project Phase 3 Identifying variables Week 4, Friday SI assignment Find the Bayesian errors of SI Week 4, Friday

Common mistake Control work 1. Memorizing the formula is not enough: define the space, define the events, check the division, then calculate.

1. Newbold, P., Carlson, W. & Thorne, B. Statistics for Business and Economics. Pearson. Chapter 3 — Conditional probability, the complete probability formula, Bayes' theorem.
2. McClave, J. T. & Sincich, T. (2018). Statistics, 13th ed. Pearson. Chapter 3.
3. Heumann, C., Schomaker, M. & Shalabh (2016). Introduction to Statistics and Data Analysis. Springer. Chapter 6.
4. Çetinkaya-Rundel, M. & Hardin, J. (2024). Introduction to Modern Statistics, 2nd ed. OpenIntro — Conditional probability and contingency tables.
5. Gigerenzer, G. & Hoffrage, U. (1995). How to improve Bayesian reasoning without instruction: frequency formats. *Psychological Review*, 102(4), 684–704.
6. Kahneman, D. & Tversky, A. (1973). On the psychology of prediction. *Psychological Review*, 80(4), 237–251—base rate error.
7. Thompson, W. C. & Schumann, E. L. (1987). Interpretation of statistical evidence in criminal trials. *Law and Human Behavior*, 11(3), 167–187 — prosecutorial error.
8. Interactive: Seeing Theory, Brown University, Department of Bayes' Theorem — seeing-theory.brown.edu